

A FRACTIONAL WEAK TILING MEASURE IN $\mathbb{Z}_{24}$ THAT IS NO CONVEX COMBINATION OF TILING COMPLEMENTS

ABSTRACT. Kadir and Fan ask, in the third item of the open problems of arXiv:2607.02149, whether every nonnegative weak tiling measure μ of a set $E \subseteq \mathbb{Z}_M$ is a convex combination of the indicator measures of classical tiling complements of E . We answer this in the negative for cyclic groups. Take $M = 24$ and $E = \{0, 4, 6, 10\}$, a set which genuinely tiles $\mathbb{Z}_{24}$ and has exactly sixteen tiling complements, and let

$$g = \mathbf{1}_{\{0,8,16\}} + \frac{1}{2} \mathbf{1}_{\{7,9,11,19,21,23\}}.$$

Then $g \geq 0$, $g(0) = 1$ and $\mathbf{1}_E * g = \mathbf{1}_{\mathbb{Z}_{24}}$, and the linear functional $L(x) = x(7) + x(9) + x(11) + x(19) + x(21) + x(23)$ takes the value 3 at g and at most 2 at each of the sixteen tiling complements, so g lies outside their convex hull. Since g satisfies the source's normalisation $f(0) = 1$ and the separation is against all sixteen complements, the object answers the question negatively both with that normalisation and without it. The same construction, applied to $E = d \cdot \{0, 2, 3, 5\}$ in $\mathbb{Z}_{12d}$, refutes the question for every modulus divisible by 12 and larger than 12.

1. THE QUESTION

Let $M \geq 2$ and let $\mathbb{Z}_M = \mathbb{Z}/M\mathbb{Z}$. For $E, T \subseteq \mathbb{Z}_M$ we write $E \oplus T = \mathbb{Z}_M$ to mean that every element of $\mathbb{Z}_M$ has exactly one representation $e + t$ with $e \in E$, $t \in T$; such a T is a *tiling complement* of E . Convolution is $(\mathbf{1}_E * f)(y) = \sum_{e \in E} f(y - e)$.

Kadir and Fan [1] work with the following notion.

Definition ([1]). A set A *weakly tiles* G if there exists a nonnegative function $f: G \rightarrow \mathbb{R}_+$ such that $f(0) = 1$ and $\mathbf{1}_A * f = \mathbf{1}_G$.

The ‘‘Open problems’’ section of [1] states three problems ‘‘for general finite groups $\mathbb{Z}_M$ ’’; the third, which carries no printed number, reads:

For a set $E \subseteq \mathbb{Z}_M$, let μ be a non-negative weak tiling measure for E , that is, $\mathbf{1}_E * \mu = \mathbf{1}_{\mathbb{Z}_M}$. Is it true that μ can be represented as a convex combination of measures associated with classical tiling complements of E ? More precisely, do there exist finitely many sets $T_1, \dots, T_N \subseteq \mathbb{Z}_M$ and coefficients $c_1, \dots, c_N \geq 0$, with $\sum_{k=1}^N c_k = 1$, such that $\mu = \sum_{k=1}^N c_k \delta_{T_k}$, where each T_k satisfies $E \oplus T_k = \mathbb{Z}_M$?

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Remark (the normalisation). The inline “that is” clause of the quoted item abbreviates the Definition and omits the normalisation $f(0) = 1$. We keep the normalisation, since it is part of the notion the source studies: without it every nonempty A weakly tiles G , by $f \equiv 1/|A|$. Accordingly all statements below concern the polytope

$$(1) \quad P_E = \{g: \mathbb{Z}_M \rightarrow \mathbb{R} : g \geq 0, \mathbf{1}_E * g = \mathbf{1}_{\mathbb{Z}_M}, g(0) = 1\},$$

and the question is whether P_E is contained in the convex hull of $\{\mathbf{1}_T : E \oplus T = \mathbb{Z}_M\}$. Note that if $\mu = \sum_k c_k \mathbf{1}_{T_k}$ with $c_k \geq 0$, $\sum_k c_k = 1$ and $\mu(0) = 1$, then $c_k = 0$ whenever $0 \notin T_k$; so under this reading only the complements through 0 can carry weight, and those satisfy $\mathbf{1}_T \in P_E$. Nothing is lost by keeping the normalisation: the measure of Theorem 2 satisfies $g(0) = 1$, and the functional exhibited there separates it from the convex hull of *all* tiling complements of E , so it answers the quoted item read either way.

The normalisation constrains the support.

Lemma 1. *Let $g \in P_E$. Then $g(e - e') = 0$ for all distinct $e, e' \in E$; that is, $\text{supp}(g) \subseteq \{0\} \cup (\mathbb{Z}_M \setminus (E - E))$.*

Proof. Take $y = e \in E$ in $\mathbf{1}_E * g = \mathbf{1}_{\mathbb{Z}_M}$: then $\sum_{e' \in E} g(e - e') = 1$. The term $e' = e$ already contributes $g(0) = 1$ and all terms are nonnegative, so every other term vanishes. $\square$

Thus, after the normalisation is imposed, P_E is exactly the fractional exact cover polytope of $\mathbb{Z}_M \setminus E$ by the translates $\{E + t\}$ with $t \in \mathbb{Z}_M \setminus (E - E)$, whose 0/1 points are precisely the tiling complements of E containing 0. The quoted item therefore asserts that this polytope is integral for every pair (M, E) , and a single non-0/1 vertex refutes it.

2. THE COUNTEREXAMPLE

Theorem 2. *Let $M = 24$, let $E = \{0, 4, 6, 10\}$, and let $g: \mathbb{Z}_{24} \rightarrow \mathbb{Q}_{\geq 0}$ be*

$$\begin{aligned} g(0) &= g(8) = g(16) = 1, \\ g(7) &= g(9) = g(11) = g(19) = g(21) = g(23) = \tfrac{1}{2}, \end{aligned}$$

and $g = 0$ at the remaining fifteen residues. Then

- (i) *E tiles $\mathbb{Z}_{24}$, and has exactly sixteen tiling complements, namely the sets $(2a + \langle 8 \rangle) \cup (2b + 1 + \langle 8 \rangle)$ with $a, b \in \{0, 1, 2, 3\}$, of which exactly four contain 0;*
- (ii) *$g \in P_E$, and g is not 0/1;*
- (iii) *g is not a convex combination of measures $\mathbf{1}_T$ with $E \oplus T = \mathbb{Z}_{24}$.*

Consequently the third open problem of [1] has a negative answer for cyclic groups.

Proof. (ii). Nonnegativity and $g(0) = 1$ are visible. For $\mathbf{1}_E * g = \mathbf{1}$, split by parity. We have

$$E - E = \{0, 2, 4, 6, 10, 14, 18, 20, 22\},$$

which is contained in the even residues, so the three translates $E + 0 = \{0, 4, 6, 10\}$, $E + 8 = \{8, 12, 14, 18\}$ and $E + 16 = \{16, 20, 22, 2\}$ are pairwise disjoint and partition the twelve even residues; each even y therefore receives exactly one contribution, of weight 1. For odd y , the six translates $E + s$ with $s \in \{7, 9, 11, 19, 21, 23\}$ cover each of the twelve odd residues exactly twice — explicitly, y is covered by $E + s$ for exactly the two $s \in \{7, 9, 11, 19, 21, 23\}$ with $y - s \in E$ — and $2 \cdot \frac{1}{2} = 1$. Since $g(7) = \frac{1}{2} \notin \{0, 1\}$, g is not 0/1.

(i). Every element of $\bar{E}$ is even, so $E + t$ consists of even residues for even t and of odd residues for odd t . Hence if $E \oplus T = \mathbb{Z}_{24}$ then the even part T_0 of T must satisfy $E \oplus T_0 = 2\mathbb{Z}_{24}$ and the odd part T_1 must satisfy $E \oplus T_1 = 1 + 2\mathbb{Z}_{24}$; in particular $|T_0| = |T_1| = 12/4 = 3$. Also $(T - T) \cap (E - E) = \{0\}$, since $t - t' = e' - e$ would give $t + e = t' + e'$. Within one parity class the available differences are even, and the even residues *not* in $E - E$ are 8, 12, 16; a 3-subset of a parity class with all pairwise differences in $\{8, 12, 16\}$ must be a coset of $\langle 8 \rangle = \{0, 8, 16\}$, because a pair at difference 12 admits no third element ($12 + 8 = 20$, $12 + 16 = 4$ and $12 + 12 = 0$ all reintroduce a forbidden difference). So $T_0 = 2a + \langle 8 \rangle$ and $T_1 = 2b + 1 + \langle 8 \rangle$ with $a, b \in \{0, 1, 2, 3\}$, which is sixteen candidates. Each of them tiles: since

$$E \oplus \langle 8 \rangle = \{0, 4, 6, 10\} \cup \{8, 12, 14, 18\} \cup \{16, 20, 22, 2\} = 2\mathbb{Z}_{24},$$

translating by $2a$ gives $E \oplus T_0 = 2\mathbb{Z}_{24}$ and translating by $2b + 1$ gives $E \oplus T_1 = 1 + 2\mathbb{Z}_{24}$. Exactly four of the sixteen contain 0, namely

$$\begin{aligned} \{0, 1, 8, 9, 16, 17\}, & \quad \{0, 3, 8, 11, 16, 19\}, \\ \{0, 5, 8, 13, 16, 21\}, & \quad \{0, 7, 8, 15, 16, 23\}. \end{aligned}$$

(iii). Put

$$L(x) = x(7) + x(9) + x(11) + x(19) + x(21) + x(23),$$

so that $L(g) = 6 \cdot \frac{1}{2} = 3$. Let $T = T_0 \cup T_1$ be one of the sixteen. The six residues 7, 9, 11, 19, 21, 23 are odd, so $L(\mathbf{1}_T) = |T_1 \cap H|$ with $H = \{7, 9, 11, 19, 21, 23\}$, and the four possibilities are

$$\begin{aligned} \{1, 9, 17\} \cap H &= \{9\}, & \{3, 11, 19\} \cap H &= \{11, 19\}, \\ \{5, 13, 21\} \cap H &= \{21\}, & \{7, 15, 23\} \cap H &= \{7, 23\}. \end{aligned}$$

Hence $L(\mathbf{1}_T) \in \{1, 2\}$ for all sixteen T , so $L \leq 2$ on their convex hull, while $L(g) = 3 > 2$. $\square$

3. EVERY MODULUS DIVISIBLE BY 12 ABOVE 12

Write $E' = \{0, 2, 3, 5\} \subseteq \mathbb{Z}_{12}$, $U = \{0, 4, 8\}$ and $V = \{3, 4, 5, 9, 10, 11\}$, all inside $\mathbb{Z}_{12}$.

Proposition 3. *Let $d \geq 1$, $M = 12d$ and $E = d \cdot E' = \{0, 2d, 3d, 5d\} \subseteq \mathbb{Z}_M$. Define $g_d: \mathbb{Z}_M \rightarrow \mathbb{Q}_{\geq 0}$ by*

$$\begin{aligned} g_d(dj) &= \mathbf{1}_U(j) \quad (j \in \mathbb{Z}_{12}), \\ g_d(dj + c) &= \frac{1}{2} \mathbf{1}_V(j) \quad (j \in \mathbb{Z}_{12}, \ 1 \leq c \leq d-1), \end{aligned}$$

and $g_d = 0$ elsewhere. Then E tiles $\mathbb{Z}_M$, $g_d \in P_E$ with total mass $3d$, and g_d is an extreme point of P_E . For $d \geq 2$ it is not $0/1$, so the third open problem of [1] fails for every $M \in \{12d : d \geq 2\}$. For $d = 2$ it is exactly the object of Theorem 2.

Proof. Let $A[y][t] = 1$ if $y - t \in E$ and 0 otherwise, $y, t \in \mathbb{Z}_M$, so that $\mathbf{1}_E * g = \mathbf{1}_{\mathbb{Z}_M}$ reads $Ag = \mathbf{1}$. Since $E \subseteq d\mathbb{Z}_M$, the equation $y - t \in E$ forces $y \equiv t \pmod{d}$; writing $y = dj + c$ and $t = ds + c'$ we get $y - t \in E$ iff $c = c'$ and $j - s \in E'$. So both the convolution and the matrix A decouple over the d cosets of $d\mathbb{Z}_M$, each block being the $\mathbb{Z}_{12}$ circulant $B[j][s] = \mathbf{1}_{E'}(j - s)$.

Tiling: $E' \oplus U = \mathbb{Z}_{12}$ (the twelve sums $e' + u$ are $0, 2, 3, 5, 4, 6, 7, 9, 8, 10, 11, 1$, all distinct), hence with $T = \{dt + c : t \in U, 0 \leq c \leq d - 1\}$ we get $E \oplus T = \{d(e' + u) + c\} = \mathbb{Z}_M$, and $0 \in T$.

Membership: on the coset $c = 0$ the block equation is $B\mathbf{1}_U = \mathbf{1}$, i.e. $E' \oplus U = \mathbb{Z}_{12}$, which was just checked; on each coset $c \neq 0$ it is $B(\frac{1}{2}\mathbf{1}_V) = \mathbf{1}$, i.e. each $j \in \mathbb{Z}_{12}$ has exactly two representations $j = e' + v$ with $e' \in E'$, $v \in V$. For that, note $V = \{3, 4, 5\} \oplus \{0, 6\}$ and that the multiset $E' + \{3, 4, 5\}$ is $\{3, 4, 5, 5, 6, 6, 7, 7, 8, 8, 9, 10\}$, whose multiplicity function m satisfies $m(j) + m(j + 6) = 2$ for every $j \in \mathbb{Z}_{12}$; the number of representations of j over V is $m(j) + m(j - 6) = m(j) + m(j + 6) = 2$. Nonnegativity and $g_d(0) = 1$ are visible, and the mass is $3 + 3(d - 1) = 3d$.

Extremality: the support columns of A are, block by block, the columns of B indexed by U in the block $c = 0$ and by V in each block $c \neq 0$. Because A is block diagonal it suffices that $B|_U$ has rank 3 and $B|_V$ has rank 6. The first holds since the three columns have disjoint supports E' , $E' + 4$, $E' + 8$; the second because the 6×6 submatrix of $B|_V$ on the rows $(6, 7, 8, 0, 1, 2)$

is block diagonal with two copies of $\begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}$ and so has determinant 4.

Hence the $3 + 6(d - 1)$ support columns of A are independent, so g_d is a basic feasible solution of $\{x \geq 0, Ax = \mathbf{1}, x(0) = 1\}$ and therefore an extreme point of P_E [5, §8.5]. For $d \geq 2$ some coset $c \neq 0$ exists and g_d takes the value $\frac{1}{2}$, so g_d equals no $\mathbf{1}_T$; by Remark 1 any convex combination $\sum_k \lambda_k \mathbf{1}_{T_k}$ equal to g_d may be assumed supported on complements through 0, all of which lie in P_E , and an extreme point is no convex combination of points of P_E other than itself. For $d = 1$ there is no coset $c \neq 0$ and $g_1 = \mathbf{1}_{\{0, 4, 8\}}$ is integral, which is why $d \geq 2$ is required. $\square$

4. SCOPE

The variants in the same list of [1] that quantify over *positive definite* measures are untouched by the object above: g is real but not symmetric ($g(7) = \frac{1}{2}$ while $g(-7) = g(17) = 0$), hence not positive definite. We do not claim that $M = 24$ is the least modulus carrying such an object; Theorem 2 and Proposition 3 say nothing about $M \notin 12\mathbb{Z}$.

The question descends from the open problems of Kolountzakis, Lev and Matolcsi [2], which ask whether every extreme point of the weak tiling body is a proper tiling; for related results in the finite abelian setting see [3, 4].

5. VERIFICATION

The accompanying program `verify.py` (Python 3.9+, standard library only, exact integer and `Fraction` arithmetic, no external data file) recomputes the twenty-four convolution values, finds the sixteen tiling complements by exhaustive enumeration of all $\binom{24}{6} = 134,596$ six-subsets of $\mathbb{Z}_{24}$, evaluates L on all of them, and checks the four complements through 0 against the list printed above. For Proposition 3 it checks the two $\mathbb{Z}_{12}$ block ranks and the members $d = 2, \dots, 8$ end to end; the block decomposition and the support pattern for general d are not computed, so the passage from those members to every $M \in \{12d : d \geq 2\}$ rests on the proof above and not on computation. The program prints one line per check and exits 0 only if all of them pass; the recorded run is `verify.output.txt`.

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